**12.** Each element of the center of a group $G$ gives rise to a 1-element conjugate class of $G$. It is clear from Exercise 8a that for $n > 2$, every permutation in $S_n$ having an orbit with more than one element is conjugate to some other permutation in $S_n$. Thus if $n > 2$, the identity in $S_n$ is the only element that is conjugate only to itself.

## 38. Free Abelian Groups

**1.** {1, 1, 1), (1, 2, 1), (1, 1, 2)} is a basis. (Note that the 2nd - 1st gives (0, 1, 0) and the 3rd - 1st gives (0, 0, 1), so it is clear that this set generates, and it has the right number of elements for a basis by Theorem 38.6.

**2.** Yes, {(2, 1), (3, 1)} is a basis. Now (1, 0) = (3, 1) + (-1)(2, 1) and (0, 1) = 3(2, 1) + (-2)(3, 1) so {(2, 1), (3, 1)} generates $\mathbb{Z} \times \mathbb{Z}$. If $m(2,1)+n(3,1)+=(0,0)$, then $2m+3n=0$ and $m+n=0$. Then $m=-n$ so $2(-n)+3n=0, n=0$, and $m=0$. Thus the conditions for a basis in Theorem 38.1 are satisfied.

**3.** (See the answer in the text.)

**4.** By Cramer's rule, the equations

$$\begin{aligned} ax + cy &= e \\ bx + dy &= f \end{aligned}$$

have a unique solution in $\mathbb{R}$ if and only if $ad - bc \neq 0$. The solution is then

$$x = \frac{ed - fc}{ad - bc} \quad \text{and} \quad y = \frac{af - be}{ad - bc}.$$

These values $x$ and $y$ are integers for *all* choices of $e$ and $f$ if and only if $D = ad - bc$ divides each of $a$, $b$, $c$, and $d$. Let $a = r_a D, b = r_b D, c = r_c D$, and $d = r_d D$. Then $D = ad - bc = (r_a r_d - r_b r_c)D^2$ so that $D$ is an integer which is an integer multiple of its square. The only possible values for $D$ are $\pm 1$, so we obtain the condition $|ad - bc| = 1$.

**5.** The definition is incorrect. Change "generating set" to "basis".

The **rank** of a free abelian group $G$ is the number of elements in a basis for $G$.

**6.** The definition is correct.

**7.** $2\mathbb{Z}$ is a proper subgroup of rank $r = 1$ of the free abelian group $\mathbb{Z}$ of rank $r = 1$.

**8.** T T T T T F F T T F

**9.** Let $\phi : G \to \underbrace{\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}}_{r \text{ factors}}$ be the map described before the statement of Theorem 38.5. Note that $\phi$ is well defined because each $a \in G$ has a *unique* expression in the form $n_1x_1 + n_2x_2 + \cdots + n_rx_r$ where each $n_i \in \mathbb{Z}$. Suppose $b \in G$, and $b = m_1x_1 + m_2x_2 + \cdots + m_rx_r$. Then

$$\begin{aligned} \phi(a + b) &= \phi[(n_1 + m_1)x_1 + (n_2 + m_2)x_2 + \cdots + (n_r + m_r)x_r] \\ &= (n_1 + m_1, n_2 + m_2, \cdots, n_r + m_r) \\ &= (n_1, n_2, \cdots, n_r) + (m_1, m_2, \cdots, m_r) \\ &= \phi(a) + \phi(b) \end{aligned}$$

so $\phi$ is a homomorphism. If $\phi(a) = \phi(b)$, then $n_i = m_i$ for $i = 1, 2, \cdots, r$ so $a = b$; this shows that $\phi$ is one to one. Clearly $\phi$ is an onto map because $n_1x_1 + n_2x_+ \cdots + n_rx_r$ is in $G$ for all integer choices of the coefficients $n_i$, for $i = 1, 2, \cdots, r$. Thus $\phi$ is an isomorphism.

**10.** Let $G$ be free abelian with a basis $X$. Let $a \neq 0$ in $G$ be given by $a = n_1x_1 + n_2x_2 + \cdots + n_rx_r$ where $x_i \in X$ and $n_i \in \mathbb{Z}$ for $i = 1, 2, \cdots, r$. If $a$ has finite order $m > 0$, then $ma = mn_1x_1 + mn_2x_2 + \cdots + mn_rx_r = 0$. Because $G$ is free abelian and $X$ is a basis, we deduce that $mn_1 = mn_2 = \cdots = mn_r = 0$, so $n_1 = n_2 = \cdots = n_r = 0$ and $a = 0$, contradicting our choice of $a$. Thus no element of $G$ has finite order $> 0$.

**11.** Suppose that $G$ and $G'$ are free abelian with bases $X$ and $X'$ respectively. Let $X = \{(x, 0) \mid x \in X\}$ and $\overline{X}\,' = \{(0, x') \mid x' \in X'\}$. We claim that $Y = \overline{X} \cup \overline{X}\,'$ is a basis for $G \times G'$. Let $(g, g') \in G \times G'$. Then

$$g = n_1x_1 + \cdots + n_rx_r \text{ and } g' = m_1x_1\,' + \cdots + m_sx_s\,'$$

for *unique* choices of $n_i$ and $m_j$, except for possible zero coefficients. Thus

$$(g, g') = n_1(x_1, 0) + \cdots + n_r(x_r, 0) + m_1(0, x_1\,') + \cdots + m_s(0, x_s\,')$$

for *unique* choices of the $n_i$ and $m_j$, except for possible zero coefficients. This shows that $Y$ is a basis for $G \times G'$, which is thus free abelian.

**12.** If $G$ is free abelian of finite rank, then $G$ is of course finitely generated, and by Exercise 10, $G$ has no elements of finite order. Conversely, if $G$ is a finitely generated torsion-free abelian group, then Theorem 11.12 shows that $G$ is isomorphic to a direct product of the group $\mathbb{Z}$ with itself a finite number of times, so $G$ is free abelian of finite rank.

**13.** Because $\mathbb{Q}$ is not cyclic, any basis for $\mathbb{Q}$ must contain at least two elements. Suppose $n/m$ and $r/s$ are in a basis for $\mathbb{Q}$ where $n, m, r$, and $s$ are nonzero integers. Then

$$mr\frac{n}{m} + (-ns)\frac{r}{s} = rn - nr = 0,$$

which is an impossible relation in a basis. Thus $\mathbb{Q}$ has no basis, so it is not a free abelian group.

**14.** Suppose $p^ra = 0$ and $p^sb = 0$. Then $p^{r+s}(a + b) = p^s(p^ra) + p^r(p^sb) = p^s0 + p^r0 = 0 + 0 = 0$, so $a + b$ is also of $p$-power order. Also $0 = p^r0 = p^r[a + (-a)] = p^ra + p^r(-a) = 0 + p^r(-a)$, so $-a$ also has $p$-power order. Thus all elements of $T$ of $p$-power order, together with zero, form a subgroup $T_p$ of $T$.

**15.** Given the decomposition in Theorem 11.12, it is clear that the elements of $T$ of p-power order are precisely those having 0 in all components except those of the form $\mathbb{Z}_{p^r}$. (Recall that the order of an element in a direct product is the least common multiple of the orders of its components in the individual groups.) Thus $T_p$ is isomorphic to the direct product of those factors having $p$-power order.

**16.** Suppose that $na = nb = 0$ for $a, b \in G$. Then $n(a + b) = na + nb = 0 + 0 = 0$. This shows that $G[n]$ is closed under the group addition. If $na = 0$, then $0 = na = n[a + (-a)] = na + n(-a) = 0 + n(-a)$, so $n(-a) = 0$ also. Of course $n0 = 0$. Thus $G[n]$ is a subgroup of $G$.

**17.** Let $x \in \mathbb{Z}_{p^r}$. If $px = 0$, then $px$, computed in $\mathbb{Z}$, is a multiple of $p^r$. The possibilities for $x$ are

$$0, 1p^{r-1}, 2p^{r-1}, 3p^{r-1}, \cdots, (p-1)p^{r-1}.$$

Clearly these elements form a subgroup of $\mathbb{Z}_{p^r}$ that is isomorphic to $\mathbb{Z}_p$.